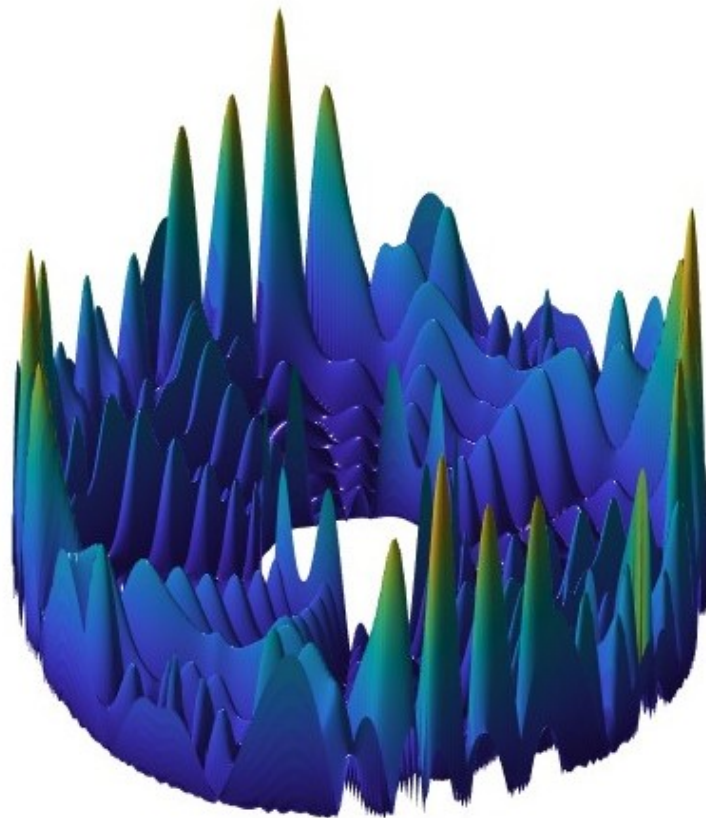




Chair of Numerical Modelling and Simulation

Isoparametric boundary conformal layers for trimmed surfaces

Semester Project



Author: Selim SHERIF

Supervisors: Pablo Antolin Sanchez, Giuliano Guarino

July 7, 2026

Contents

1	Introduction	2
2	Domain Setup and Mappings	2
3	Spline-Based Approximation of the Composite Map	3
4	$h - p$ Convergence Study	4
4.1	Geometry Choice	4
4.2	Error Quantification	5
4.3	Integral Convergence Study	6
4.4	The $h-p$ Convergence Study	6
4.4.1	Algorithm And Setup	6
4.4.2	Results	7
4.4.3	Interpretation & Comments	11
5	Static Analysis Study	11
5.1	Problem Description	11
5.2	Mesh and Spline Configuration	11
5.3	Results & Observations	12
6	Supplement: Spatial Modes and Visual Errors	13
7	Conclusion	17
	References	17

1 Introduction

Spline surfaces are a foundational tool in computer-aided design (CAD) and numerical simulation due to their ability to represent complex geometries with high smoothness and control. In particular, B-spline and NURBS mappings offer structured, parametric representations well-suited for geometric modeling, mesh generation, and isogeometric analysis.

In industrial and scientific applications, geometric models often include nontrivial features such as holes, notches, and inclusions. These are typically handled using the *trimming approach*, where a surface is first defined over a regular parametric domain and then restricted to a subset defined by trimming curves. While trimming provides flexibility, it introduces geometric and numerical challenges. The trimmed region generally does not align with the underlying spline parametrization, complicating integration, meshing, and simulation workflows.

To address these limitations, recent methods such as the *Immersed Boundary Conformal Method* (IBCM) [3] enrich the trimmed region with additional geometric layers modeled via *compositions of spline maps*. These composite mappings take the form

$$\mathcal{C}_i = \widehat{\mathcal{F}} \circ \widetilde{\mathcal{F}}_i,$$

where $\widetilde{\mathcal{F}}_i$ defines a local deformation and $\widehat{\mathcal{F}}$ maps to the physical surface. While this offers increased geometric fidelity, it comes at the cost of computational complexity: composed mappings are not themselves splines, making them more difficult to evaluate and integrate efficiently.

The central objective of this work is to investigate whether such composite spline mappings can be approximated by a single, standard spline

$$\widetilde{\mathcal{C}}_i \approx \mathcal{C}_i,$$

thus simplifying geometry representations while preserving sufficient accuracy for downstream simulation tasks. This approximation raises several key questions:

- How does the approximation error behave as a function of spline degree p and knot spacing h ?
- What qualitative trends emerge across different geometric configurations and refinement strategies?
- To what extent does the approximation affect simulation outcomes, such as those from linear static analyses?

This study aims to quantify the trade-offs between geometric fidelity and computational tractability in the context of composed spline geometries arising from trimming and conformal layer generation.

2 Domain Setup and Mappings

The construction of the final surface geometry relies on a sequence of mappings between three distinct spaces: auxiliary domains, a structured parametric domain, and the physical space. This hierarchical setup enables flexible trimming and boundary layer definition, while preserving a structured spline-based representation suitable for analysis.

We begin with the reference parametric domain

$$\widehat{\Pi}_0 = [\xi_1^b, \xi_1^t] \times [\xi_2^b, \xi_2^t] \subset \mathbb{R}^2, \quad (1)$$

where the superscripts b and t denote the lower and upper bounds in each parametric direction, respectively. This is mapped to the physical space through a spline-based geometric transformation,

$$\widehat{\mathcal{F}} : \widehat{\Pi}_0 \rightarrow \Pi_0 \subset \mathbb{E}^3, \quad \mathbf{x} = \widehat{\mathcal{F}}(\xi_1, \xi_2). \quad (2)$$

The degree of the spline defining the mapping $\widehat{\mathcal{F}}$ is denoted by $\widehat{p}$. Here, Π_0 denotes the full untrimmed surface in physical space. A trimmed physical surface $\Omega_0 \subset \Pi_0$ is defined by restricting the parametric domain to a trimmed region $\widehat{\Omega}_0 \subset \widehat{\Pi}_0$, such that:

$$\Omega_0 = \widehat{\mathcal{F}}(\widehat{\Omega}_0), \quad \partial\Omega_0 = \widehat{\mathcal{F}}(\partial\widehat{\Omega}_0). \quad (3)$$

To define boundary layers around internal and external features of a trimmed surface, a collection of *auxiliary parametric domains* $\widetilde{\Omega}_i \subset \mathbb{R}^2$ is introduced, one for each layer $i = 1, 2, \dots, n$. Each domain is defined as a simple rectangular region:

$$\widetilde{\Omega}_i = [\eta_1^{i,b}, \eta_1^{i,t}] \times [\eta_2^{i,b}, \eta_2^{i,t}], \quad (4)$$

Each auxiliary domain is then mapped into the trimmed parametric domain $\widehat{\Pi}_0$ via a curvilinear transformation:

$$\widetilde{\mathcal{F}}_i : \widetilde{\Omega}_i \rightarrow \widehat{\Omega}_i \subset \widehat{\Pi}_0. \quad (5)$$

The map $\widetilde{\mathcal{F}}_i$ is also defined by a spline representation, whose degree is denoted by $\widetilde{p}$. Finally, the corresponding *physical boundary layers* Ω_i are obtained by composing the local and global mappings, yielding the composite transformation $\mathcal{C}_i := \widehat{\mathcal{F}} \circ \widetilde{\mathcal{F}}_i$, such that

$$\Omega_i = \mathcal{C}_i(\widetilde{\Omega}_i), \quad i = 1, 2, \dots, n. \quad (6)$$

A schematic representation of this multi-domain construction is provided in Figure 1, illustrating the sequential mapping from auxiliary grids to the final surface geometry in physical space.

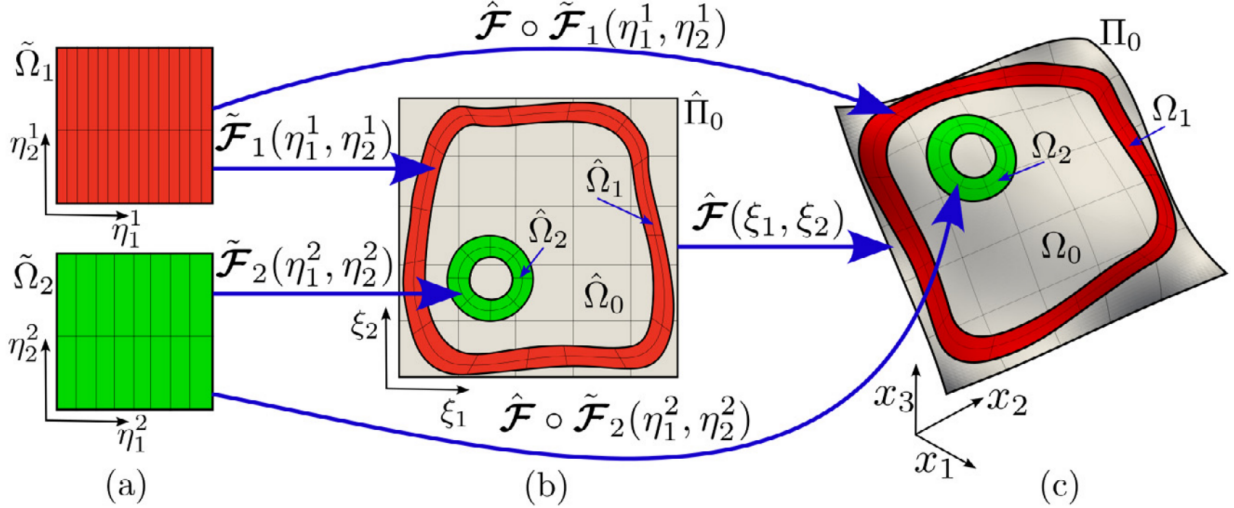


Figure 1 | Mapping pipeline from structured auxiliary domains to the physical surface. (a) Multiple auxiliary domains $\tilde{\Omega}_i \subset \mathbb{R}^2$, typically defined as simple rectangles, serve as local parametrizations for boundary layer regions. (b) These are mapped via curvilinear transformations $\tilde{\mathcal{F}}_i$ into corresponding trimmed regions $\hat{\Omega}_i \subset \hat{\Pi}_0$, the global parametric domain. (c) The global mapping $\hat{\mathcal{F}}$ then lifts both the interior region $\hat{\Omega}_0$ and the boundary layers $\hat{\Omega}_i$ into the physical domain $\Pi_0 \subset \mathbb{R}^3$, producing a composite surface consisting of the main geometry Ω_0 and its associated boundary layers Ω_i . Image credit: [1]

Overall, The geometric construction involves three distinct categories of coordinate spaces, each associated with specific domains, local variables, and transformation maps. While only one global parametric domain and one physical domain exist, multiple auxiliary domains are typically introduced, one for each boundary feature or layer.

Auxiliary domains $\tilde{\Omega}_i \subset \mathbb{R}^2$: simple rectangular patches indexed by $i = 1, \dots, n$, used to define structured parameterizations of boundary layer regions. Each auxiliary domain is mapped into the parametric domain via a curvilinear transformation $\tilde{\mathcal{F}}_i : \tilde{\Omega}_i \rightarrow \hat{\Omega}_i \subset \hat{\Pi}_0$. All quantities in this space are marked with a tilde, and local coordinates are denoted by (η_1^i, η_2^i) .

Parametric domain $\hat{\Pi}_0 \subset \mathbb{R}^2$: the global reference domain which includes the trimmed region $\hat{\Omega}_0$, and the boundary layers $\hat{\Omega}_i$. The map from the parametric to the physical space is given by $\hat{\mathcal{F}} : \hat{\Pi}_0 \rightarrow \Pi_0$. Quantities in this domain carry a hat, and the coordinates are denoted by (ξ_1, ξ_2) .

Physical domain $\Pi_0 \subset \mathbb{R}^3$: the embedded surface in physical space, composed of the main trimmed region Ω_0 and the boundary layers Ω_i . Each physical boundary layer is obtained via composition of the two maps as $\Omega_i = \mathcal{C}_i(\tilde{\Omega}_i)$. Physical-space quantities are left unadorned and use Cartesian coordinates (x_1, x_2, x_3) .

3 Spline-Based Approximation of the Composite Map

With the above framework in mind, the task is to approximate the composite map $\mathcal{C}_i = \hat{\mathcal{F}} \circ \tilde{\mathcal{F}}_i$, which maps the auxiliary domain $\tilde{\Omega}_i \subset \mathbb{R}^2$ into the physical surface $\Pi_0 \subset \mathbb{R}^3$. Although both $\hat{\mathcal{F}}$ and $\tilde{\mathcal{F}}_i$ are spline maps, their composition $\mathcal{C}_i$ is, in general, not a spline due to the nonlinear nature of function composition. This lack of spline structure makes direct manipulation and integration of $\mathcal{C}_i$ both analytically intractable and computationally expensive. In contrast, spline functions offer a highly practical and efficient framework for numerical computation, geometry representation, and integration. To overcome these difficulties, the goal is to construct a degree- p NURBS approximation $\tilde{\mathcal{C}}_i \in [\mathcal{S}_p]^3$ that best approximates $\mathcal{C}_i$ in the L^2 -norm. Here, $[\mathcal{S}_p]^3$ denotes a vector-valued spline space, where each of the three physical coordinates (x_1, x_2, x_3) is represented by an independent function in the same scalar spline space $\mathcal{S}_p$.

Let $\{N_k(\boldsymbol{\eta})\}_{k=1}^N$ be a chosen spline basis on $\tilde{\Omega}_i$. The approximating spline map is then expressed as

$$\tilde{\mathcal{C}}_i(\boldsymbol{\eta}) = \sum_{k=1}^N \mathbf{X}_k N_k(\boldsymbol{\eta}) \in \mathbb{R}^3, \quad (7)$$

where the control-point vectors $\mathbf{X}_k \in \mathbb{R}^3$ are computed by minimizing the squared L^2 -error:

$$\min_{\{\mathbf{X}_k\}} \int_{\tilde{\Omega}_i} \left\| \mathcal{C}_i(\boldsymbol{\eta}) - \sum_{k=1}^N \mathbf{X}_k N_k(\boldsymbol{\eta}) \right\|^2 d\boldsymbol{\eta}. \quad (8)$$

This minimization leads to the normal equations

$$\sum_{k=1}^N \left(\int_{\tilde{\Omega}_i} N_j(\boldsymbol{\eta}) N_k(\boldsymbol{\eta}) d\boldsymbol{\eta} \right) \mathbf{X}_k = \int_{\tilde{\Omega}_i} N_j(\boldsymbol{\eta}) \mathcal{C}_i(\boldsymbol{\eta}) d\boldsymbol{\eta}, \quad (9)$$

for all $j = 1, \dots, N$. The solution yields the unique L^2 -projection $\tilde{\mathcal{C}}_i$, which is the closest degree- p spline representation of $\mathcal{C}_i$ in the chosen function space. The corresponding approximation algorithm is summarized below:

Algorithm 1: Approximation Algorithm for $\mathcal{C}_i$

- Input** : Composite map $\mathcal{C}_i : \tilde{\Omega}_i \rightarrow \mathbb{R}^3$,
NURBS structure for Ω_i
- Output**: Approximated spline map $\tilde{\mathcal{C}}_i$
- 1 Extract spline space $[\mathcal{S}_p]^3$ from the NURBS structure on $\tilde{\Omega}_i$
 - 2 Evaluate basis functions $\{N_k(\tilde{\boldsymbol{\eta}})\}$ at quadrature nodes in $\tilde{\Omega}_i$
 - 3 Assemble mass matrix $M_{kl} = \int_{\tilde{\Omega}_i} N_k N_l d\tilde{\boldsymbol{\eta}}$
 - 4 Evaluate $\mathcal{C}_i(\tilde{\boldsymbol{\eta}})$ at quadrature nodes
 - 5 Assemble RHS vector $V_k = \int_{\tilde{\Omega}_i} \mathcal{C}_i N_k d\tilde{\boldsymbol{\eta}}$
 - 6 Solve $M\mathbf{X} = V$ for control points $\{\mathbf{X}_k\}$
 - 7 Construct approximation $\tilde{\mathcal{C}}_i(\tilde{\boldsymbol{\eta}}) = \sum_k \mathbf{X}_k N_k(\tilde{\boldsymbol{\eta}})$
-

4 $h - p$ Convergence Study

4.1 Geometry Choice

The untrimmed geometry Π_0 (see Fig. 2-a) used in the convergence study is a saddle-shaped surface, reminiscent of a Pringle chip. It is defined analytically by the height function

$$z(x, y) = (x - 0.5)^2 - (y - 0.5)^2, \quad (x, y) \in [0, 1]^2. \quad (10)$$

This surface offers a balanced combination of regularity and geometric variation: it is neither overly symmetric nor artificially complex. As such, it serves as a compelling benchmark for evaluating the fidelity of spline-based approximations. The moderate curvature challenges the approximation space while avoiding numerical artifacts that may arise in more intricate geometries.

In practical applications, the geometry of the surface is typically dictated by the underlying engineering or scientific problem and must be approximated as faithfully as possible. In such cases, the analyst has little or no control over the shape. In the present study, however, a well-defined saddle-shaped surface is deliberately selected to enable a controlled investigation. The knot vector defining the surface is held fixed, while the spline degree of the NURBS discretization is systematically varied. This setup preserves the global geometry and enables a focused assessment of how the polynomial degree

influences approximation quality, independently of geometric complexity.

More precisely, the reference surface Π_0 is defined by a NURBS mapping with spline degree $\hat{p} \in \{3, 4, 5\}$ in both parametric directions ξ_1 and ξ_2 , using a fixed internal knot vector $\{0.25, 0.5, 0.75\}$ in each direction. Varying the degree $\hat{p}$ directly influences the regularity and smoothness of the basis functions. In particular, the surface exhibits global continuity of order $C^{\hat{p}-1}$ in the absence of repeated knots, making $\hat{p}$ a key parameter for assessing how continuity affects approximation quality.

By contrast, altering the knot vector would affect the locality and spacing of the basis functions, modifying the distribution of control points and slightly perturbing the geometry. However, such changes primarily influence local shape and do not qualitatively alter convergence behavior. Therefore, to isolate the influence of spline degree alone, the knot vector of the reference surface map is held fixed across all cases. This provides a consistent geometric foundation for controlled comparisons and avoids introducing spurious variations due to local reparameterization effects.

Building on this fixed background geometry, localized refinement is introduced through a trimmed annular region centered at the minimum of the saddle surface, effectively modeling a geometric exclusion at that location. This construction marks the point at which composition is introduced into the framework. The refinement region is defined over a rectangular auxiliary domain $\tilde{\Omega}_i \subset \mathbb{R}^2$, equipped with local coordinates (η_1, η_2) , and is mapped into the global parametric domain $\hat{\Pi}_0$ via the previously defined transformation $\tilde{\mathcal{F}}_i$. The resulting trimmed patch,

$$\hat{\Omega}_i = \tilde{\mathcal{F}}_i(\tilde{\Omega}_i), \quad (11)$$

is centered at $(0.5, 0.5)$ and bounded by inner and outer radii $r_{\text{in}} = 0.10$ and $r_{\text{out}} = 0.35$, respectively (see Fig. 2-b). Within this region, the auxiliary spline degree $\tilde{p}$ and mesh size h of $\tilde{\mathcal{F}}_i$ are systematically varied to evaluate their impact on the approximation error and convergence behavior in the composed geometry. The corresponding physical patch is obtained by composing with the global surface mapping $\hat{\mathcal{F}}$, yielding

$$\mathcal{C}_i := \hat{\mathcal{F}} \circ \tilde{\mathcal{F}}_i, \quad \Omega_i = \mathcal{C}_i(\tilde{\Omega}_i) \subset \mathbb{R}^3, \quad (12)$$

which defines the target geometry used in the convergence analysis. (see Fig. 2-c)

Note. Given the anisotropic nature of the annular region, the mesh is designed to preserve the aspect ratio between radial and circumferential elements. Specifically, approximately **6 circumferential elements** (i.e., roughly 2π) are used per **radial element**, aligning with the natural scaling of arc length with radius. This ensures that the grid cells maintain nearly equilateral proportions in the physical domain.

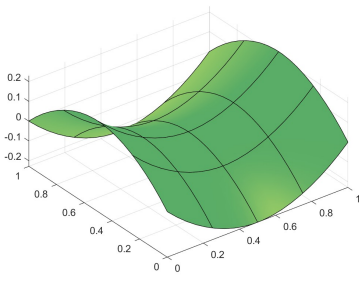
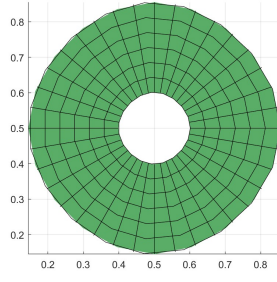
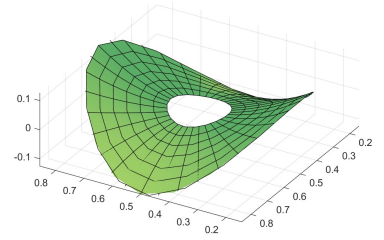
(a) Saddle surface Ω_0 (b) Trimmed parametric region $\hat{\Omega}_1$ (c) Physical boundary layer Ω_1

Figure 2 | Three-stage domain mapping for constructing a boundary layer. (a) The underlying surface $\Omega_0 \subset \Pi_0$ is defined via the spline map $\hat{\mathcal{F}}$ from the parametric domain $\hat{\Pi}_0$. (b) A trimmed patch $\hat{\Omega}_1 \subset \hat{\Pi}_0$ is obtained by mapping a rectangular auxiliary domain $\tilde{\Omega}_1 \subset \mathbb{R}^2$ using $\tilde{\mathcal{F}}_1$. (c) The physical layer $\Omega_1 = \hat{\mathcal{F}} \circ \tilde{\mathcal{F}}_1(\tilde{\Omega}_1)$ forms a curved annulus embedded in the saddle surface.

4.2 Error Quantification

To assess the convergence of the approximated surface with respect to the exact composed geometry, we define several error metrics. These are grouped into three main categories:

(1) L^2 Pointwise Errors. This error is evaluated at each point on the discretized surface and quantifies the local discrepancy between the exact geometry $\mathcal{C}_i$ and its spline approximation $\tilde{\mathcal{C}}_i$. The pointwise L^2 error is defined as:

$$e_{L^2}(\eta_1, \eta_2) = \sqrt{(x_{\text{ex}} - \tilde{x})^2 + (y_{\text{ex}} - \tilde{y})^2 + (z_{\text{ex}} - \tilde{z})^2} \quad (13)$$

(2) Integral Errors. To globally quantify the quality of the approximation, we define the following relative L^2 -based metrics integrated over the surface:

(a) Geometry Error (Full):

$$e_{L^2}^{\text{Full}} = \frac{\int_{\Omega} \|\mathbf{x} - \mathbf{x}_{\text{ex}}\|^2 d\Omega}{\int_{\Omega} \|\mathbf{x}_{\text{ex}}\|^2 d\Omega} \quad (14)$$

(b) Normal Vector Error (Full):

$$e_N^{\text{Full}} = \frac{\int_{\Omega} \|\mathbf{n} - \mathbf{n}_{\text{ex}}\|^2 d\Omega}{\int_{\Omega} \|\mathbf{n}_{\text{ex}}\|^2 d\Omega} \quad (15)$$

where $\mathbf{n}$ and $\mathbf{n}_{\text{ex}}$ are the unit normals of the approximate and exact surfaces, respectively.

(c) H^1 Seminorm Error:

$$e_{H^1} := \frac{\int_{\Omega} \left\| \frac{\partial(\mathbf{x} - \mathbf{x}_{\text{ex}})}{\partial \xi_1} \right\|^2 + \left\| \frac{\partial(\mathbf{x} - \mathbf{x}_{\text{ex}})}{\partial \xi_2} \right\|^2 d\Omega}{\int_{\Omega} \left\| \frac{\partial \mathbf{x}_{\text{ex}}}{\partial \xi_1} \right\|^2 + \left\| \frac{\partial \mathbf{x}_{\text{ex}}}{\partial \xi_2} \right\|^2 d\Omega} \quad (16)$$

(3) Boundary Errors. To assess the fidelity of the spline approximation along the trimmed annulus, we define relative L^2 errors integrated along each boundary curve $\Gamma_{\star} \in \{\Gamma_{\text{In}}, \Gamma_{\text{Out}}\}$, where $\star = \{\text{In}, \text{Out}\}$. These boundaries are illustrated in Figure 3.

(a) Geometry Error:

$$e_{L^2}^{\star} = \frac{\int_{\Gamma_{\star}} \|\mathbf{x} - \mathbf{x}_{\text{ex}}\|^2 d\Gamma}{\int_{\Gamma_{\star}} \|\mathbf{x}_{\text{ex}}\|^2 d\Gamma} \quad (17)$$

(b) Normal Vector Error:

$$e_n^{\star} = \frac{\int_{\Gamma_{\star}} \|\mathbf{n} - \mathbf{n}_{\text{ex}}\|^2 d\Gamma}{\int_{\Gamma_{\star}} \|\mathbf{n}_{\text{ex}}\|^2 d\Gamma} \quad (18)$$

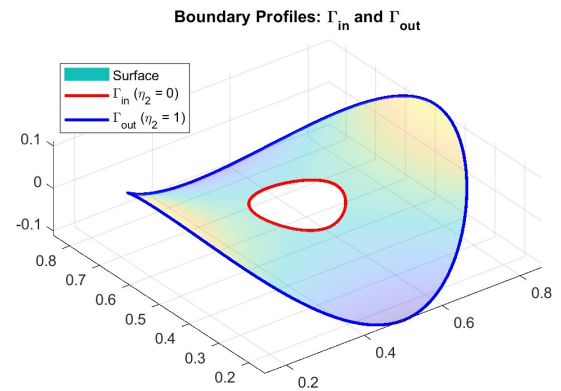


Figure 3 | Annular domain Ω_1 . The inner boundary Γ_{in} , shown in red, and the outer boundary Γ_{out} , shown in blue, are defined by the trimming curves used to construct the domain.

4.3 Integral Convergence Study

Before performing the h - p convergence analysis, an *integral convergence study* is conducted to verify that numerical surface integrals are evaluated with sufficient accuracy. This step ensures that the reported errors are not dominated by quadrature artifacts.

All integration is performed directly on the physical surface Ω using a structured evaluation grid. The surface integral of a scalar field q is approximated by a weighted sum:

$$\int_{\Omega} q dA \approx \sum_{i,j} w_{ij} J_{ij} q_{ij}, \quad (19)$$

where:

- q_{ij} is the integrand evaluated at grid point (i, j) ,
- $J_{ij} = \left\| \frac{\partial \mathbf{x}}{\partial \eta_1} \times \frac{\partial \mathbf{x}}{\partial \eta_2} \right\|$ is the Jacobian determinant of the mapping at that point,
- w_{ij} are trapezoidal quadrature weights defined by:

$$w_{ij} = \begin{cases} 1 & \text{if } (i, j) \text{ is an interior point,} \\ \frac{1}{2} & \text{if } (i, j) \text{ lies on an edge,} \\ \frac{1}{4} & \text{if } (i, j) \text{ is a corner point.} \end{cases}$$

This quadrature rule is employed to evaluate both surface and boundary integrals for all error metrics, across a range of grid resolutions N . The goal is to determine the resolution at which the integration becomes sufficiently stable to support reliable convergence studies.

Algorithm 2: Integral Convergence Study

Input : Grid levels $N_1 < \dots < N_K$; exact composite map $\mathcal{C}_i$; approximated map $\tilde{\mathcal{C}}_i$

Output: Errors e_{L_2} , $e_{L_2}^*$, e_N , e_N^* , $e_{D\mathbf{x}}$, where $\star \in \{\text{IN}, \text{OUT}\}$

```

1 for each grid level  $N$  do
2   Set mesh size  $h \leftarrow 1/N$ 
3   Generate evaluation grid on  $\tilde{\Omega}_i$  of size  $(N+1) \times (N+1)$ 
4   Evaluate maps  $\mathcal{C}_i(\boldsymbol{\eta})$  and  $\tilde{\mathcal{C}}_i(\boldsymbol{\eta})$ 
5   Evaluate surface derivatives and compute Jacobian determinant  $J(\boldsymbol{\eta})$ 
6   Construct trapezoidal quadrature weights  $w_{ij}$ 
7   Compute all error metrics:  $e_{L_2}$ ,  $e_{L_2}^*$ ,  $e_N$ ,  $e_N^*$ ,  $e_{D\mathbf{x}}$ 
8 end
9 Plot all error quantities versus  $N$  on a log-log scale

```

The integration procedure is tested on a representative configuration in which the boundary layer geometry is defined by the *composite mapping* $\mathcal{C}$. In this setup, the auxiliary map $\tilde{\mathcal{F}}$ is constructed

as a spline of degree $p = 5$ over a uniform knot vector with internal knots spaced at intervals of 0.1, i.e., $\{0.1, 0.2, \dots, 0.9\}$, while the base mapping $\hat{\mathcal{F}}$ is fixed as previously defined. This configuration is not chosen for its geometric complexity, but rather to assess the accuracy and stability of the numerical quadrature. Since the goal of this study is to verify the convergence of surface integrals under the composite geometry, the particular construction of $\tilde{\mathcal{F}}$ is of secondary importance. What matters is that the integral approximations stabilize with increasing grid resolution.

Figure 4 presents the evolution of all integral-based error metrics as a function of grid refinement. A convergence plateau is observed at $N = 128$, which is adopted for all subsequent h - p convergence studies to ensure that numerical integration error remains negligible relative to geometric approximation errors.

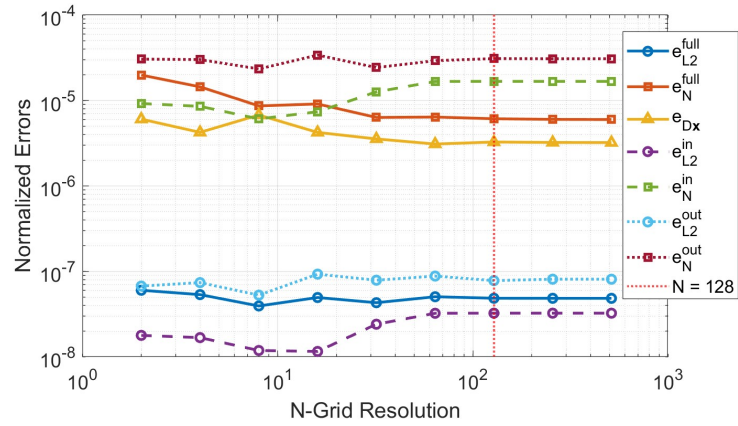


Figure 4 | Convergence of numerical integration errors as a function of grid resolution. All evaluated metrics exhibit saturation beyond $N = 128$, indicating that integration accuracy is no longer an influential factor.

4.4 The h - p Convergence Study

4.4.1 Algorithm And Setup

With the integration accuracy validated in the previous section, the next step is to assess the approximation quality of the numerical spline maps under combined h and $\tilde{p}$ -refinements. The aim is to evaluate how the geometry-induced error behaves as the polynomial degree $\tilde{p}$ and the mesh size h of the auxiliary map $\tilde{\mathcal{F}}_i$ are varied.

For each degree $\tilde{p}$, a family of auxiliary maps $\tilde{\mathcal{F}}$ is generated using knot vectors with uniform spacing determined by the mesh size h . These maps are composed with the global surface mapping to form the composite geometry $\mathcal{C} = \hat{\mathcal{F}} \circ \tilde{\mathcal{F}}$, and are subsequently projected onto the corresponding spline space, yielding the numerical approximation $\tilde{\mathcal{C}}$. The resulting approximations $\tilde{\mathcal{C}}$ are then compared against the exact composite geometry $\mathcal{C}$. Errors are evaluated on a fixed high-resolution grid of size 128×128 , as described in the preceding analysis.

Algorithm 3: h - p Convergence Study

Input : Degrees $\{\tilde{p}_1, \dots, \tilde{p}_P\}$, grid spacings $\{h_1, \dots, h_R\}$, exact geometry map $\mathcal{C}$

Output: Errors e_{L^2} , $e_{L^2}^*$, e_N , e_N^* , $e_{D\mathbf{x}}$, with $\star \in \{\text{IN}, \text{OUT}\}$

- 1 Set evaluation grid with resolution $N = 128$, and precompute quadrature weights and Jacobian
- 2 **for** each degree $\tilde{p}$ **do**
- 3 **for** each grid spacing h **do**
- 4 Build auxiliary map $\tilde{\mathcal{F}}$ of degree $\tilde{p}$ with uniform knot spacing h
- 5 **Perform Algorithm 1 to obtain projection of $\mathcal{C}$ onto spline space**
- 6 Evaluate exact and approximate surfaces over the evaluation grid
- 7 Compute derivatives and Jacobians $J(\eta_1, \eta_2)$
- 8 Compute surface normal vectors for both maps
- 9 Compute errors e_{L^2} , $e_{L^2}^*$, e_N , e_N^* , $e_{D\mathbf{x}}$
- 10 **end**
- 11 **end**
- 12 Plot errors against h on a log-log scale for each p

To assess the approximation behavior, the convergence procedure is executed independently for each of the three reference saddle-shaped surfaces defined by $\hat{p} \in \{3, 4, 5\}$. For each such base geometry, the auxiliary spline degree $\tilde{p} \in \{3, 4, 5\}$ and mesh size h are varied systematically. Refinement levels follow a dyadic sequence: $h = 1/8, 1/16, 1/32, \dots, 1/256$. Although finer resolutions were initially considered, they were excluded due to computational constraints. The resulting convergence curves, computed using the previously defined error metrics, are presented in Figures 5–8.

Table 1 summarizes the estimated convergence rates for the full-domain L^2 , normal vector, and H^1 seminorm errors. These rates are extracted using a linear fit on the last two data points. The convergence rate is computed on a log-log scale as:

$$\text{Rate} = -\frac{\log e_n - \log e_{n-2}}{\log h_n - \log h_{n-2}}, \quad (20)$$

where e_i denotes the error and h_i the corresponding mesh size at refinement level i .

These empirical results are evaluated against theoretical expectations drawn from spline approximation theory. Specifically, for a NURBS mapping of degree p , it is well established (cf. [1, 2]) that the geometric approximation error in the L^2 norm converges at a rate of $\mathcal{O}(h^{p+1})$, while the error in the unit normal vector, also measured in the L^2 norm, converges at a rate of $\mathcal{O}(h^p)$. These rates arise from the smoothness properties of spline spaces: a spline

of degree p is C^∞ within each element but only C^{p-1} continuous across element interfaces, often referred to as “knot lines” or “lines of reduced continuity.” This limited inter-element regularity constrains the differentiability of the composed geometry and thus dictates the attainable convergence order.

In the present setting, the geometry is defined by the composition of two spline mappings: a base mapping of degree $\hat{p}$, and an auxiliary mapping of degree $\tilde{p}$. The regularity of the composed map is governed by the minimum regularity of the two components. As continuity is reduced across patches, the effective differentiability of the resulting surface is given by

$$p^* - 1 = \min(\hat{p} - 1, \tilde{p} - 1), \quad (21)$$

which reflects the limiting smoothness induced by inter-patch continuity. Consequently, the expected convergence order is $p^* + 1$ for the L^2 error, and p^* for both the normal vector and H^1 seminorm errors. These expected values are indicated in gray in Table 1.

It is important to note that convergence behaviors associated with the inner and outer boundary errors (both for the L^2 norm and normal vector) are not reported in this table. These quantities exhibit irregular behavior with respect to h , indicating potential local effects or geometric irregularities that prevent a consistent asymptotic trend.

4.4.2 Results

Error Metric	$\tilde{p} = 3$	$\tilde{p} = 4$	$\tilde{p} = 5$
Underlying Surface Degree $\hat{p} = 3$			
$e_{L^2}^{\text{full}}$	3.84 (4)	3.24 (4)	3.67 (4)
e_N^{full}	2.70 (3)	2.40 (3)	2.66 (3)
e_{H^1}	2.67 (3)	2.38 (3)	2.53 (3)
Underlying Surface Degree $\hat{p} = 4$			
$e_{L^2}^{\text{full}}$	3.95 (4)	4.57 (5)	4.41 (5)
e_N^{full}	3.11 (3)	3.90 (4)	3.53 (4)
e_{H^1}	3.12 (3)	3.93 (4)	3.40 (4)
Underlying Surface Degree $\hat{p} = 5$			
$e_{L^2}^{\text{full}}$	3.95 (4)	5.34 (5)	5.72 (6)
e_N^{full}	3.11 (3)	3.94 (4)	4.73 (5)
e_{H^1}	3.11 (3)	3.94 (4)	4.51 (5)

Table 1: Estimated convergence rates for full-domain L^2 , normal vector, and H^1 seminorm errors. Expected orders are shown in parentheses based on the auxiliary degree $\tilde{p}$.

Error Metrics Convergence for $\hat{p} = 3$

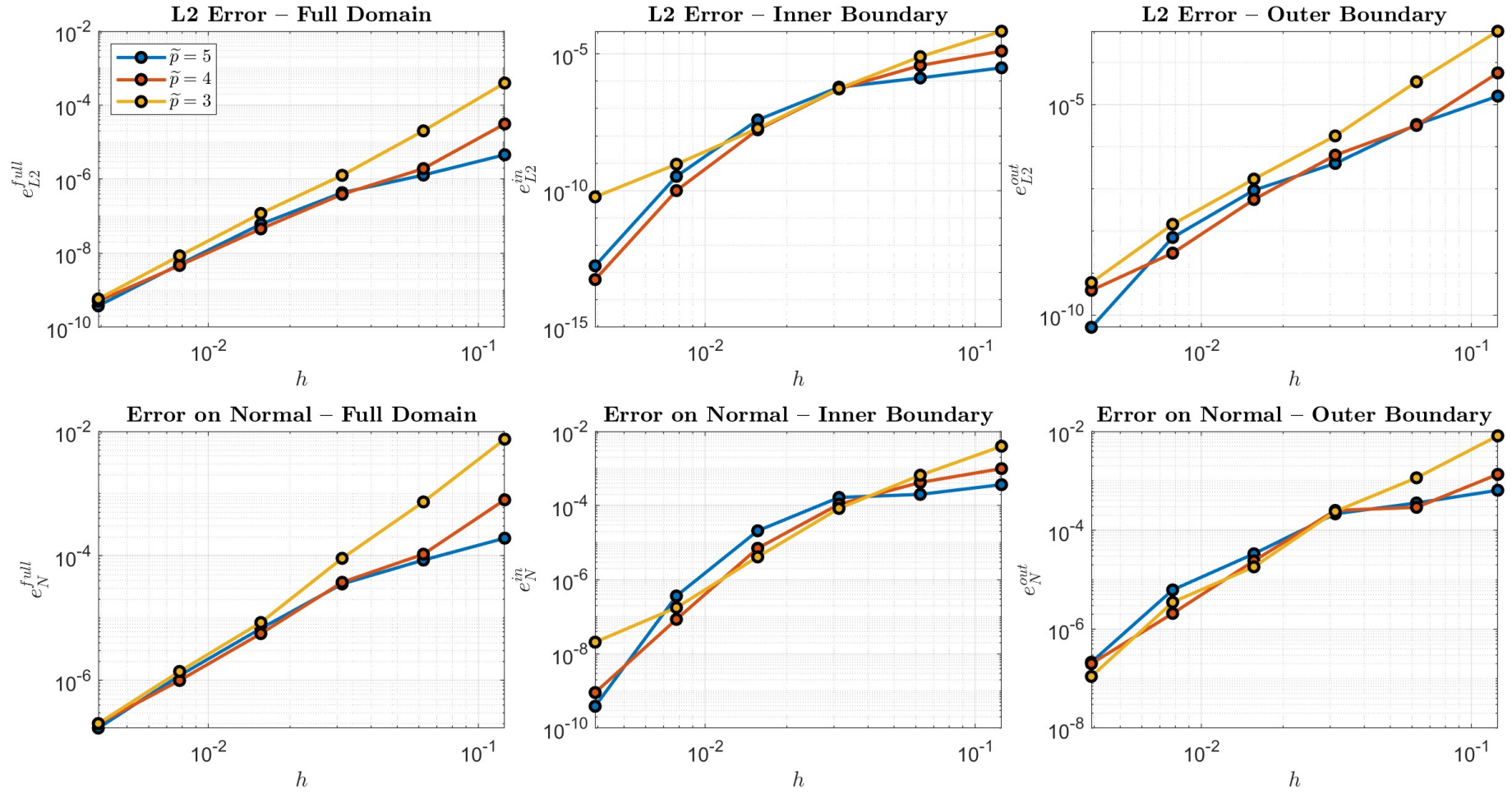


Figure 5 | Error metrics convergence for base spline degree $\hat{p} = 3$. Each subplot shows how the respective error metric varies with auxiliary refinement parameters $\tilde{p}$ and h .

Error Metrics Convergence for $\hat{p} = 4$

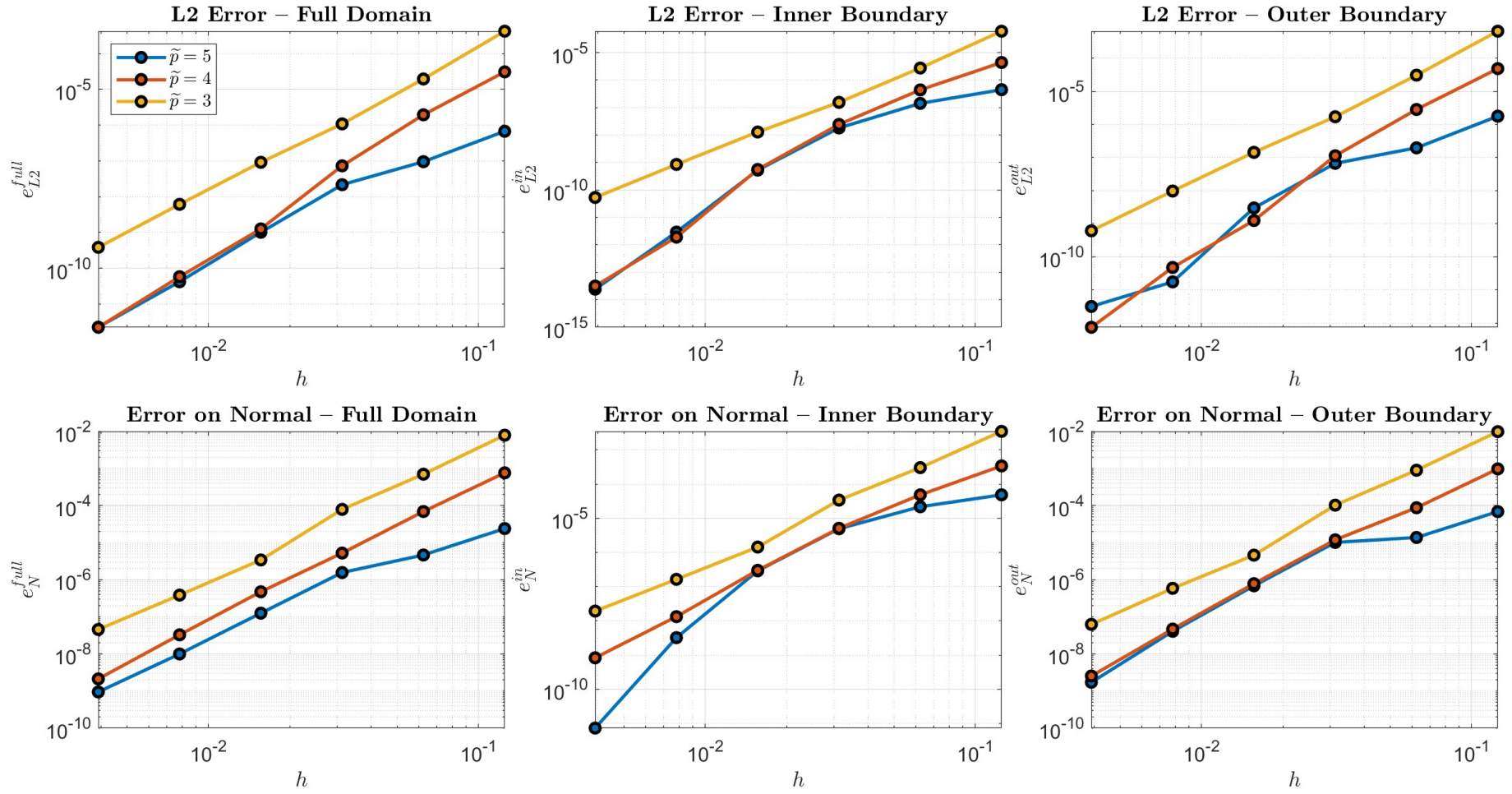


Figure 6 | Error metrics convergence for base spline degree $\hat{p} = 4$. Each subplot shows how the respective error metric varies with auxiliary refinement parameters $\tilde{p}$ and h .

Error Metrics Convergence for $\hat{p} = 5$

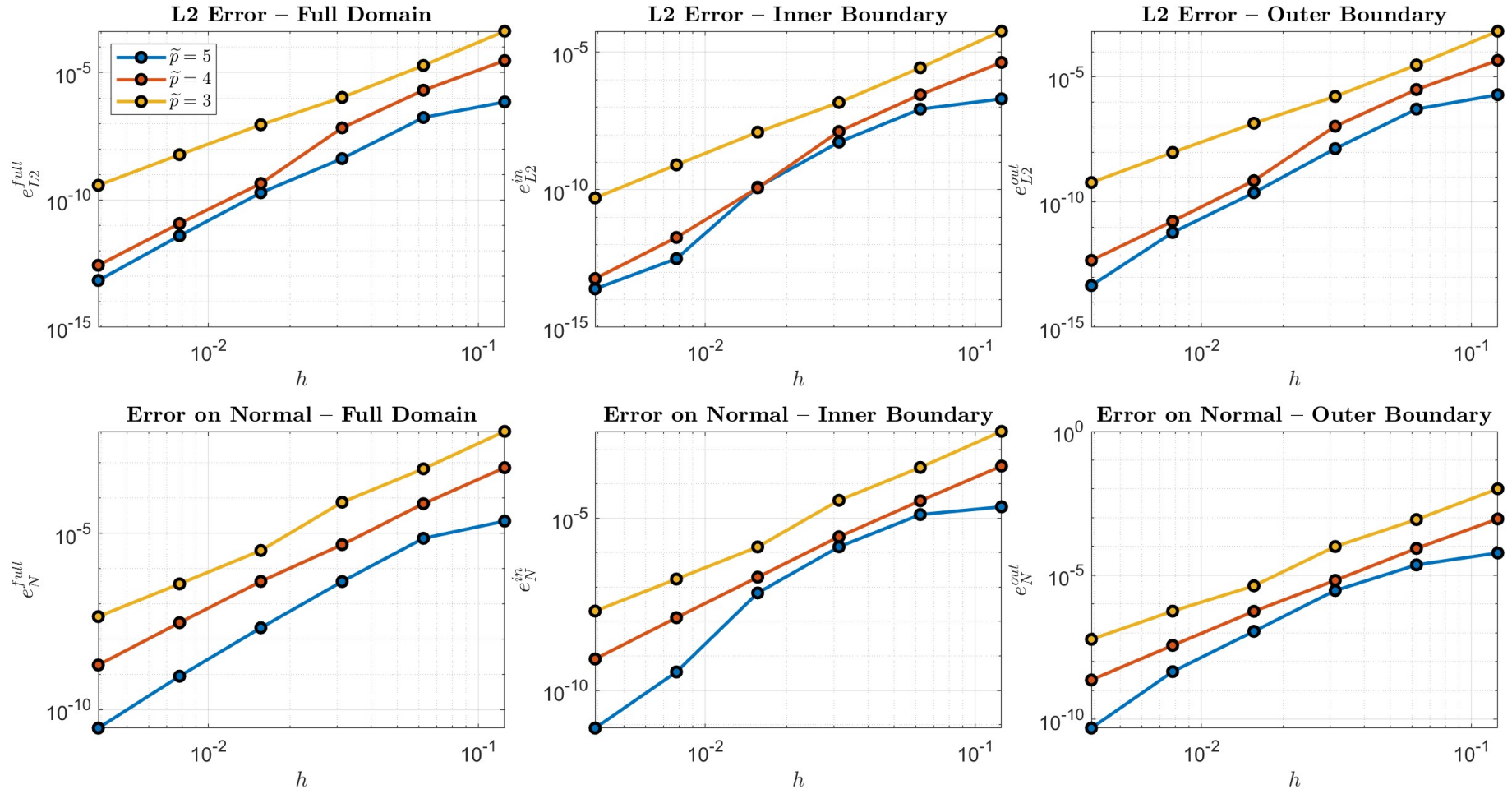


Figure 7 | Error metrics convergence for base spline degree $\hat{p} = 5$. Each subplot shows how the respective error metric varies with auxiliary refinement parameters $\tilde{p}$ and h .

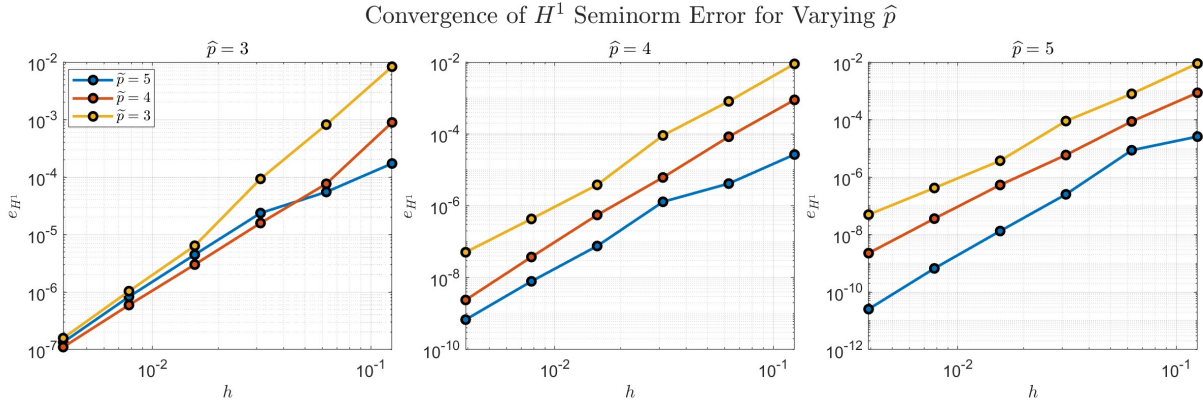


Figure 8 | Convergence for the H^1 seminorm across base spline degrees $\hat{p} = 3, 4, 5$. Each subplot shows how the derivative-based error varies with auxiliary refinement parameters $\tilde{p}$ and h .

4.4.3 Interpretation & Comments

Agreement with theoretical rates: Strong agreement is observed between empirical and expected convergence rates. Most results closely match theoretical predictions based on the minimum-regularity argument, supporting the idea that the composed map behaves like a single spline with effective smoothness p^* , defined as the minimum of the two spline regularities.

Subtle underestimation: Among the 27 estimated convergence rates, 23 fall slightly below theoretical expectations, while only 4 exceed them marginally. Although the discrepancy is minor, it reflects a consistent underestimation trend, possibly due to geometric or numerical factors not fully accounted for in the idealized regularity model. Nevertheless, the overall trend remains aligned with theoretical predictions.

Error decay and spline degree: The absolute error consistently decreases with increasing $\hat{p}$ or $\tilde{p}$, confirming that both degrees contribute positively to approximation quality, independent of which map is refined.

Irregularity in boundary errors: Boundary errors often deviate from linear log-log behavior. In particular, $\tilde{p} < \hat{p}$ configurations show concave decay, while $\tilde{p} \geq \hat{p}$ cases are more linear. These irregularities may result from local geometric distortion or reduced smoothness near trim boundaries.

5 Static Analysis Study

To assess the practical impact of geometric approximation on simulation accuracy, a static structural analysis is performed on the Pringle-shaped base surface Ω_0 , enriched with a trimmed annular refinement layer Ω_1 . The simulation is carried out using two geometric representations of this trimmed region:

- The exact composed geometry $\Omega_1 = \mathcal{C}(\tilde{\Omega}_1) =$

$$\hat{\mathcal{F}} \circ \tilde{\mathcal{F}}(\tilde{\Omega}_1),$$

- Its spline-based approximation $\tilde{\Omega}_1 = \tilde{\mathcal{C}}(\tilde{\Omega}_1)$, where $\tilde{\mathcal{C}} \in [\mathcal{S}_{\tilde{p}}]^3$ is obtained via L^2 -projection of $\mathcal{C}$.

5.1 Problem Description

A symmetric boundary displacement is applied on the top and bottom edges of the square domain. The structure is clamped and subjected to opposing displacements along the x -axis, resulting in a static deformation. This setup is designed to assess the sensitivity of the displacement field to geometric perturbations by comparing the approximate and composed mappings of the trimmed region Ω_1 .

Parameter	Value
Material model	Linear isotropic elastic
Young's modulus	$E = 1$
Poisson's ratio	$\nu = 0.25$
Shell thickness	$t = 0.05$
Dirichlet condition on left edge	$u_x = -1, u_y = 0, u_z = 0,$
Dirichlet condition on right edge	$u_x = +1, u_y = 0, u_z = 0,$

Table 2: Material properties and prescribed Dirichlet boundary conditions used in the static shell analysis.

5.2 Mesh and Spline Configuration

Two levels of discretization are considered to evaluate the influence of mesh resolution on simulation outcomes. In both cases, the global surface map is defined with spline degree $\hat{p} = 3$, and the auxiliary trimming map also uses $\tilde{p} = 3$, as modifying the polynomial degree is not the focus of this study. Instead, refinement is performed by adjusting the element sizes in each domain: we denote by h the characteristic mesh size of the base (global) surface,

and by $\tilde{h}$ the mesh size of the auxiliary trimming layer. These quantities describe the approximate physical diameter of elements in each region.

The refinement strategy ensures that $\hat{h}$ and $\tilde{h}$ are chosen to produce comparable element sizes across the composed geometry. This matching minimizes interpolation mismatch at the interface and reduces integration artifacts due to scale inconsistency between overlapping regions.

- **Coarse mesh:**

The base surface is discretized with $\hat{h} = 0.25$, while the auxiliary trimming map uses anisotropic spacing $\tilde{h} = (0.1, 0.5)$, corresponding to the azimuthal and radial directions, respectively.

- **Fine mesh:**

The base domain is refined to $\hat{h} = 1/20$, and the auxiliary domain is matched using $\tilde{h} = (1/36, 1/7)$, again for the azimuthal and radial directions. These values are selected to produce comparable element diameters throughout the composed geometry.

Mesh	$\hat{p}$	$\hat{h}$	$\tilde{p}$	$\tilde{h}$
Coarse	3	0.25	3	(0.1, 0.5)
Fine	3	1/20	3	(1/36, 1/7)

Table 3: Spline degrees and mesh spacings used for the base ($\hat{\cdot}$) and auxiliary ($\tilde{\cdot}$) geometries in the static analysis study.

For each mesh configuration, the following quantities are computed:

- The displacement field for the composed geometry, $\Omega_1 = \mathcal{C}(\tilde{\Omega}_1)$,
- The displacement field for the projected geometry, $\tilde{\Omega}_1 = \tilde{\mathcal{C}}(\tilde{\Omega}_1)$,
- The pointwise difference in displacement magnitudes, $\|\mathbf{u}_{\text{proj}}\| - \|\mathbf{u}_{\text{comp}}\|$.

5.3 Results & Observations

The simulation results for both mesh resolutions are presented and discussed in the subsequent figures (Figures 9 and 10), where the spatial distribution and magnitude of the projection-induced differences are visualized.

Observations: The results demonstrate that the projection-induced geometric approximation has a negligible impact on the outcomes of the static analysis. Even under deliberately coarse discretization (with a much rougher mesh than typically used in practice) the displacement differences remain small and spatially smooth. These discrepancies are not discernible without explicit visualization of the difference field, with deviations on the order of 10^{-3} . On the finer mesh, which reflects realistic engineering resolution, the discrepancy reduces further, with a maximum difference around 3×10^{-6} , effectively becoming imperceptible in almost all practical applications.

These findings confirm the robustness of the spline projection framework for trimmed surface modeling. It enables simplified geometric representations that preserve simulation accuracy while offering substantial advantages in computational efficiency and implementation flexibility.

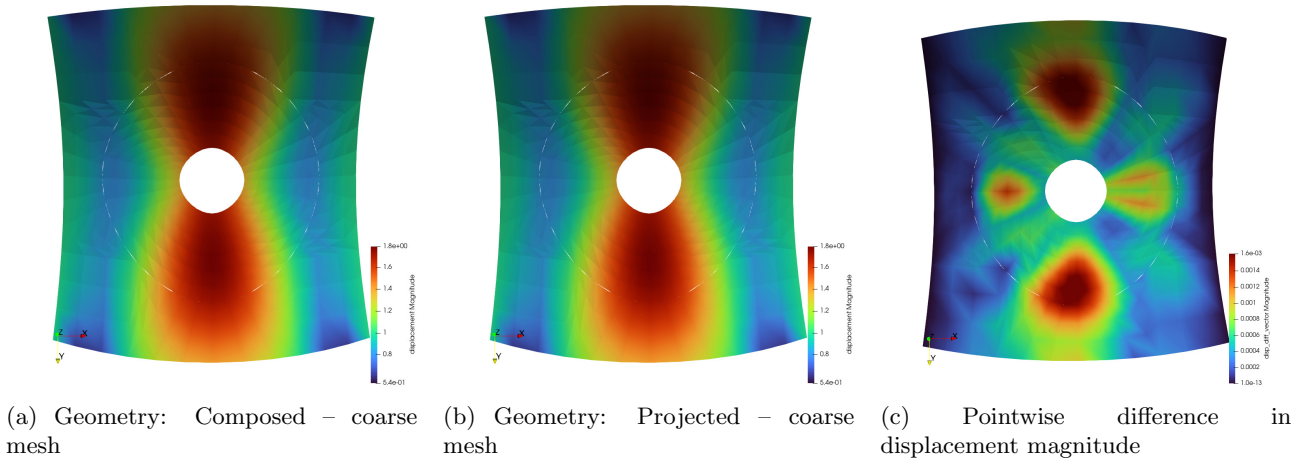


Figure 9 | Static analysis results on a **coarse mesh**. Subplots (a) and (b) show the displacement magnitude $\|\mathbf{u}\|$ for the composed and projected geometries, respectively. Subplot (c) presents the pointwise difference between these magnitudes, $\|\mathbf{u}_{\text{proj}}\| - \|\mathbf{u}_{\text{comp}}\|$. The residual reaches up to 1.6×10^{-3} .

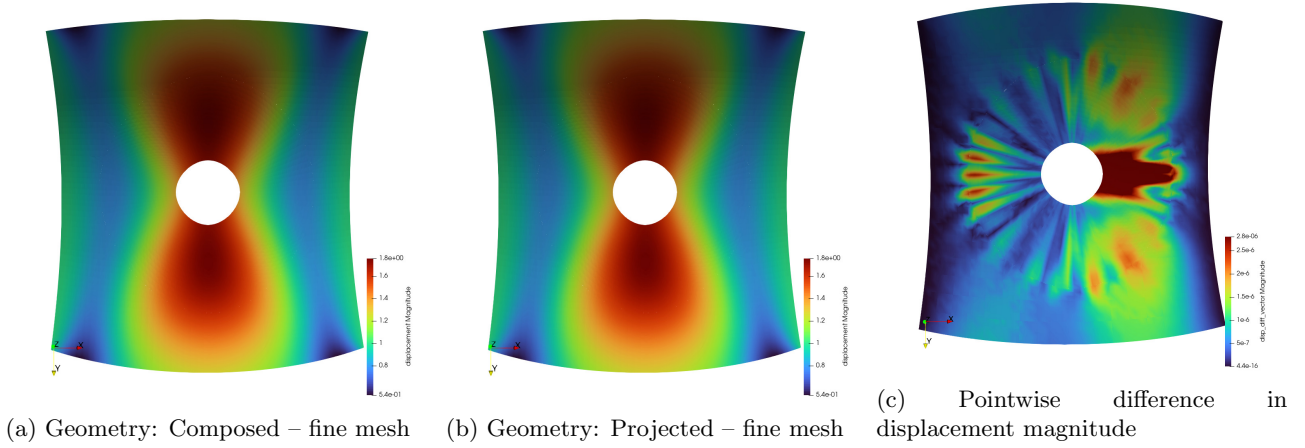


Figure 10 | Static analysis results on a **fine mesh**. Displacement magnitudes in subplots (a) and (b) are visually indistinguishable. The difference field in (c), $\|\mathbf{u}_{\text{proj}}\| - \|\mathbf{u}_{\text{comp}}\|$, drops below 10^{-6} , confirming the high-fidelity nature of the projection and its effectiveness for accurate isogeometric simulation.

6 Supplement: Spatial Modes and Visual Errors

While previous sections focused on global convergence behavior, this supplementary analysis shifts attention to the spatial organization of the pointwise error. The objective is to identify recurring visual patterns and examine their relationship to the spline degrees and refinement levels used in the projection process. **Although the results in this section are primarily visual and informal, some recurring structures appear across configurations, suggesting potential links to the underlying approximation setup. This section is therefore included as a complement to the main analysis: not to offer conclusive statements, but to provide visual context and possible avenues for interpretation.**

The pointwise L^2 error introduced in Section 4.2 is evaluated on a uniform grid of resolution $n_{\text{grid}} = 128$. For each base surface degree $\hat{p} \in \{3, 4, 5\}$, a total of 18 configurations are tested by varying the auxiliary degree $\tilde{p} \in \{3, 4, 5\}$ and knot refinement level $n \in \{8, 16, 32, 64, 128, 256\}$, with $n = 1/h$. The knot vector for $\hat{\mathcal{F}}$ is held fixed at $\{0.25, 0.5, 0.75\}$.

Figures 11–13 illustrate a variety of rich structures in the error distribution. Among these, two dominant spatial patterns consistently emerge, which are referred to as Mode 1 and Mode 2. A few intermediate cases exhibit mixed traits, though no distinct third mode is identified.

Mode 1 (Square Framing). Characterized by a rectangular band of elevated error near the domain boundary, this mode is most prominent when $\tilde{p} \geq \hat{p}$, particularly under fine refinement. A representative example is $\hat{p} = 3$, $\tilde{p} = 5$, $n = 256$, where the error is minimal in the interior and concentrated along the trimmed edge. This suggests that over-resolving the auxiliary space sharpens boundary detail but introduces framing artifacts.

Mode 2 (Radial Ripples and Bands). When $\tilde{p} < \hat{p}$, the error exhibits persistent radial oscillations or concentric bands. These patterns, as seen in cases like $\hat{p} = 4$, $\tilde{p} = 3$, $n = 256$, remain under refinement and hint at a saturation effect — a limiting configuration shaped by the mismatch in spline approximation capacity.

Mixed Features. Some configurations, such as $\hat{p} = 5$, $\tilde{p} = 4$, $n = 256$, display both square-like outer framing and faint radial patterns. These cases likely represent a transition between the dominant modes, rather than a separate regime.

Correlation with Spline Degrees. The modal behavior is strongly tied to the relative spline degrees. Mode 1 arises when the auxiliary degree is equal to or exceeds the base, offering greater local flexibility. In contrast, Mode 2 appears when the auxiliary map is underpowered, amplifying structured projection error.

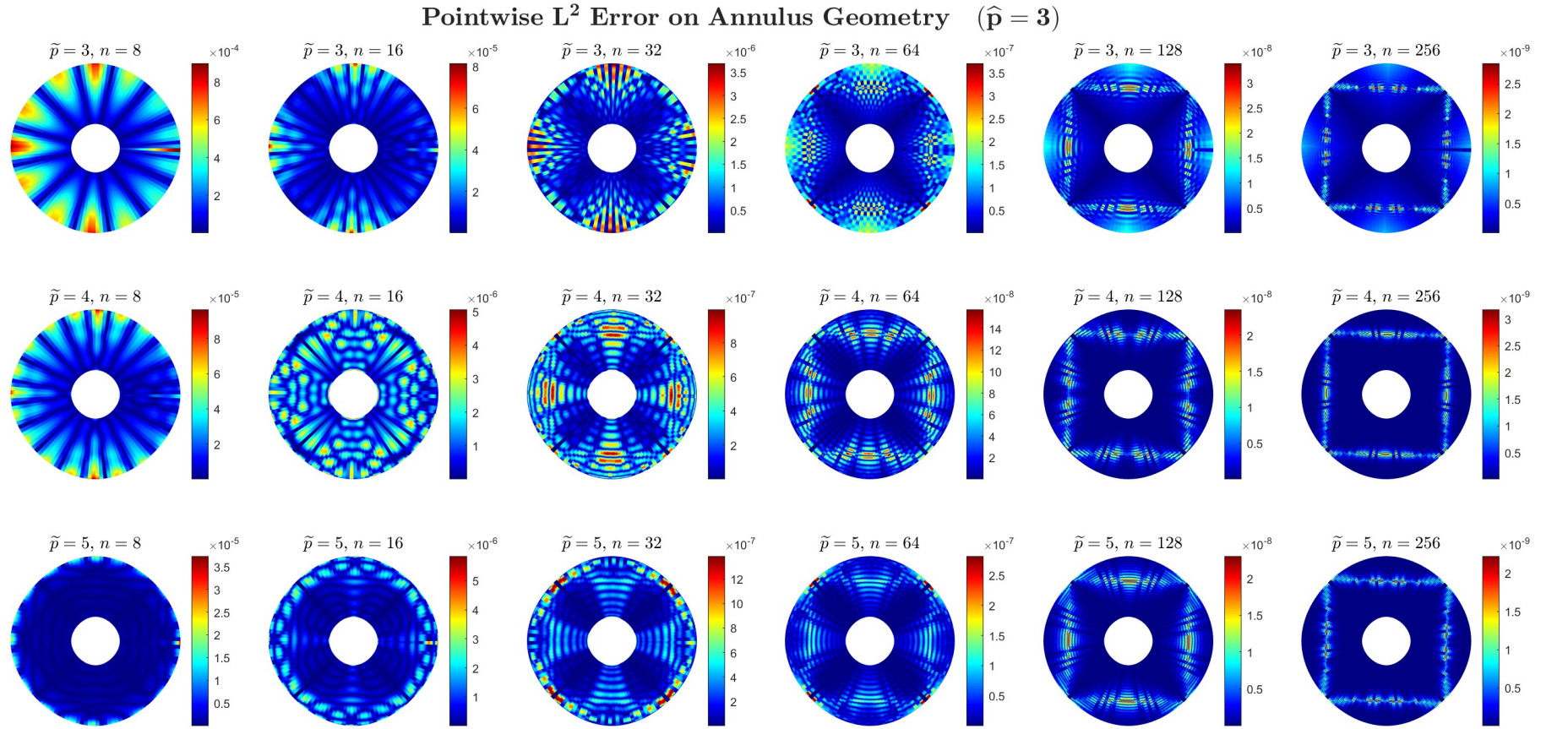


Figure 11 | Pointwise L^2 error colormaps for $\hat{p} = 3$. Configurations with $\tilde{p} \geq \hat{p}$ show clear Mode 1 structure, especially for large n .

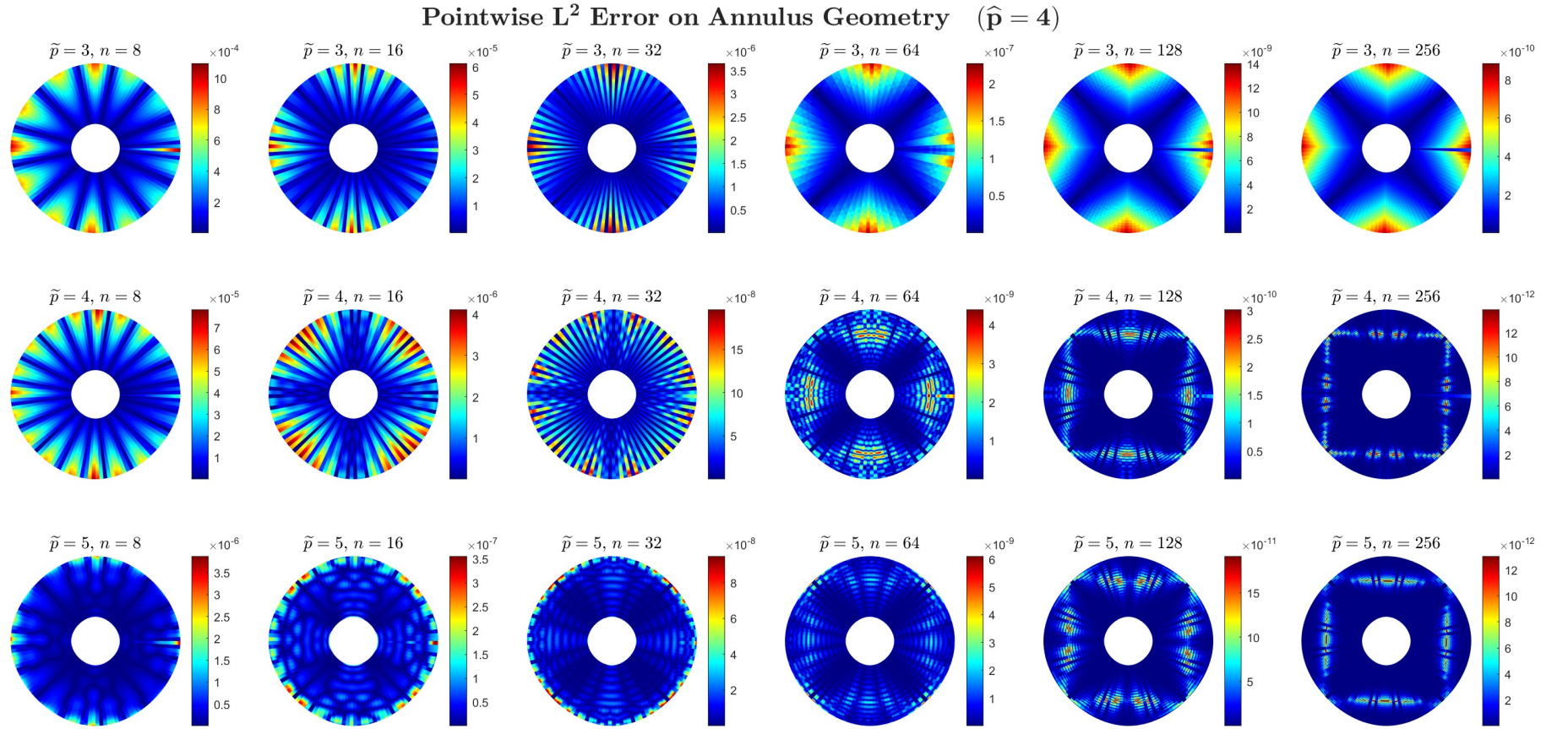


Figure 12 | Colormaps for $\hat{p} = 4$. Mode 2 dominates when $\tilde{p} < \hat{p}$; Mode 1 appears as degrees align.

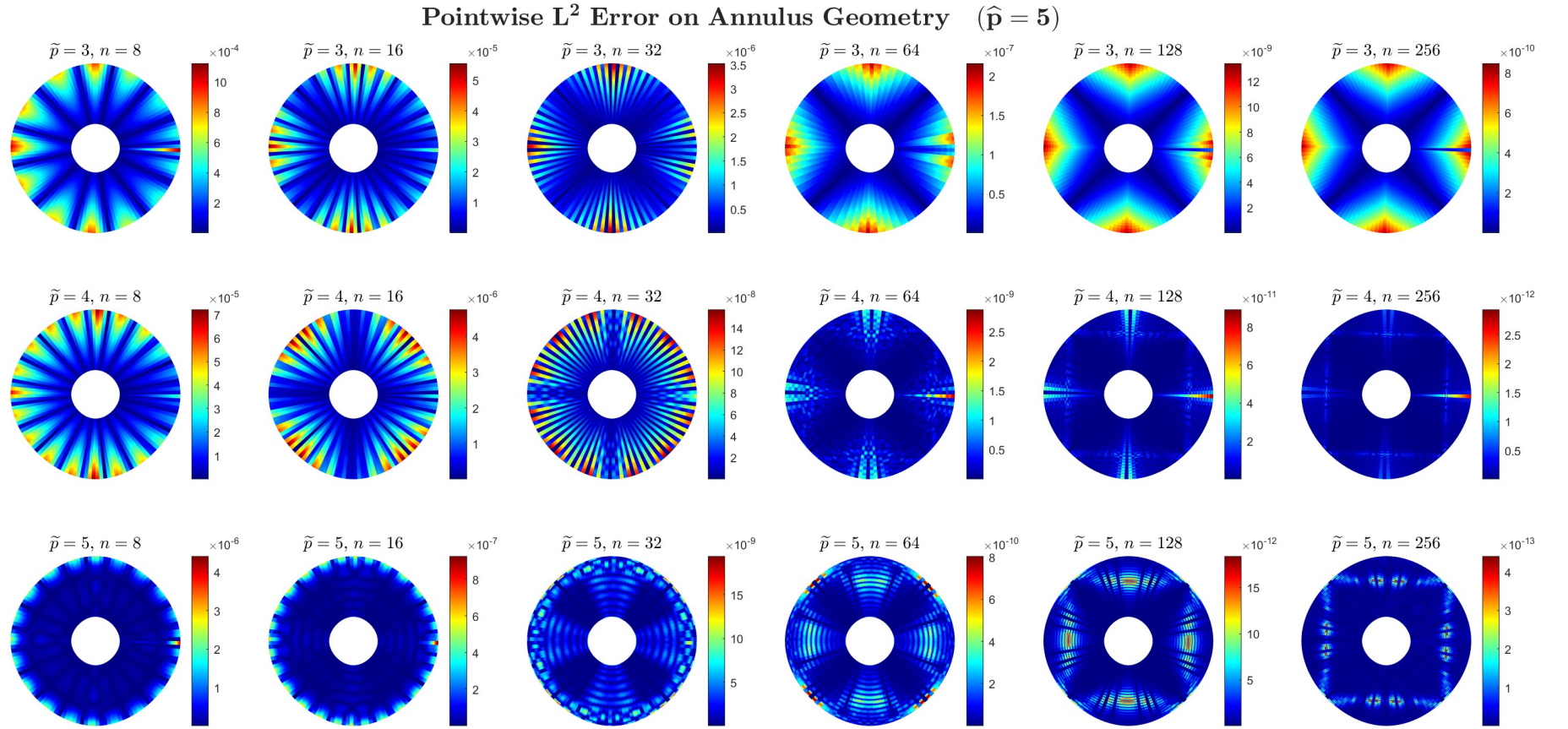


Figure 13 | Plot Error maps for $\hat{p} = 5$. Mode blending occurs in $\tilde{p} = 4$, $n = 256$; Mode 1 re-emerges for $\tilde{p} = 5$.

7 Conclusion

This study offers a detailed investigation into the approximation error arising from the projection of composed spline mappings onto target spline spaces. Both global convergence behavior and the spatial structure of the pointwise error were analyzed, providing complementary perspectives on the projection's accuracy and limitations.

Empirical convergence rates broadly follow the theoretical estimate based on the reduced regularity $p^* = \min(\hat{p}, \tilde{p})$, which characterizes the smoothness bottleneck in the composition. However, observed rates are often slightly below this ideal, likely due to geometric complexity and nonlinear interactions between component mappings—effects not fully captured by the p^* -based criterion.

Boundary errors deviate more substantially from the global trends. While full domain convergence remains relatively uniform, boundary regions exhibit irregular behavior that varies case-by-case. These discrepancies appear to stem from the geometric intricacies introduced by trimming, as well as the inherent limitations of projection in representing sharply defined boundary features.

Pointwise error visualizations support this interpretation, revealing two dominant spatial patterns which could perhaps explain the case dependant behaviour.

Importantly, despite these variations, static analysis reveals that the overall impact of the approximation remains small in practice. This underscores the practical robustness of the spline projection method within geometric modeling workflows, particularly in contexts where moderate approximation accuracy is sufficient. For applications in CAD, finite element analysis, or fluid simulations,

the method offers a viable and flexible tool, provided that ultra-high precision is not a critical constraint.

Future Directions

- **Extension to more complex geometries.** This study focuses on relatively simple annular domains. Applying the same projection framework to more intricate or irregular trimmed surfaces could uncover new error behaviors and reveal limitations of the current methodology.
- **Classification of error modes.** While two dominant error structures were consistently observed, it is unclear whether these exhaust all possible configurations. A more systematic exploration may reveal a richer taxonomy of modes or transitional behaviors, potentially influenced by the geometry of the base surface itself.
- **Beyond the p^* heuristic.** The observed convergence behavior generally aligns with the reduced regularity model $p^* = \min(\hat{p}, \tilde{p})$, but deviations suggest that additional geometric or analytical effects may be involved. Understanding these higher-order contributions could lead to a more refined theoretical model for composed spline approximation.
- **Handling geometric singularities.** Introducing kinks or sharp features into the base domain poses additional challenges for both the projection process and the resulting error structure. Investigating how such singularities affect convergence and whether new solution behaviors emerge would be an important direction for future study.

References

- [1] Antolin Pablo Guarino Giuliano and Buffa Annalisa. The immersed boundary conformal method for kirchhoff-love and reissner-mindlin shells. *Computer Methods in Applied Mechanics and Engineering*, 2024.
- [2] Thomas J.R. Hughes, J. Austin Cottrell, and Yuri Bazilevs. Isogeometric analysis: CAD, finite elements, NURBS, exact geometry and mesh refinement. *Computer Methods in Applied Mechanics and Engineering*, 2005.
- [3] Xiaodong Wei, Benjamin Marussig, Pablo Antolin, and Annalisa Buffa. Immersed boundary-conformal isogeometric method for linear elliptic problems. *Computer Methods in Applied Mechanics and Engineering*, 420, 2021.